Winter Internship Program 2026

INTERNSHIP TASK #4

Offensive Security - Information Gathering & Enumeration Report

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1. Objective

The purpose of this assignment is to develop a practical understanding of passive reconnaissance by applying both theoretical concepts and hands-on OSINT techniques learned in class. Students will simulate the role of a **junior penetration tester** performing passive information gathering on a target organization using only **open-source intelligence (OSINT)** methods. This assignment is designed to strengthen:

- Analytical thinking
- OSINT investigation skills
- Understanding of the Cyber Kill Chain (Reconnaissance phase)
- Real-world penetration testing methodology
- Technical documentation and reporting skills

2. Scenario

You have been hired as a **junior penetration tester** by a cybersecurity consultancy firm. Your firm has received a request from a client to perform **external reconnaissance** before conducting a full security assessment. Your task is to collect as much **actionable intelligence** as possible about the target **without directly interacting with their infrastructure**. You must behave like an ethical hacker working in the **reconnaissance phase** of a professional engagement.

3. Target Selection (Student Choice)

Each student must select **one legal target** for this assignment. The target must be:

- A company, university, or public organization
- Legally accessible via public internet
- **No active scanning, exploitation, or intrusion allowed**
- **Strictly passive reconnaissance only**

4. Tasks to Perform

4.1. Task 1 - Understanding the Reconnaissance Phase (Theory)

Explain in your own words:

- What is reconnaissance?
- Difference between Passive and Active reconnaissance.
- Why reconnaissance is critical in the Cyber Kill Chain.

- How attackers misuse this phase in real-world attacks.
- How ethical hackers use it defensively.

4.2. Task 2 - Google Dorking Investigation

Use advanced Google search operators to gather intelligence about the target. You must attempt to find:

- Public documents (PDF, DOC, XLS, PPT)
- Login / signup portals
- Email formats (patterns)
- Subdomains
- Exposed directories (open indexing)
- Publicly available sensitive information (if any)

Explain:

- Which dorks you used
- Why you used them
- What information you discovered

4.3. Task 3 - Metadata Analysis (YouTube / Documents)

Identify any:

- Official YouTube videos of the organization
- Public documents (PDF, DOCX, etc.) related to the organization

Extract intelligence such as:

- Author names
- Software used to create the file
- Internal naming conventions
- Possible usernames
- Email addresses

Explain how metadata can help an attacker in the reconnaissance phase.

4.4. Task 4 - Shodan Intelligence Gathering

Search the organization using **Shodan** (or alternative if no data exists). Collect information like:

- Public IP address details
- Open ports (list at least 5-10)
- Running services (HTTP, HTTPS, cPanel, WHM, webmail, etc.)
- Technologies in use (CDN, web servers, frameworks)
- Possible exposed systems

Explain:

- What each discovered port typically means
- How attackers use this data for further exploitation
- Link to **OSI layer and port theory**

4.5. Task 5 - Netcraft Analysis

Use **Netcraft** (netcraft.com) to identify:

- Hosting provider
- Web technologies (CDN, SSL, etc.)
- Server type (if detectable)
- Operating system guess (based on behaviour)
- Security headers (SPF, DMARC, DNSSEC)

Explain how this information helps in **mapping the attack surface**.

4.6. Task 6 - FTP Intelligence (NAPALM FTP Indexer)

Use **NAPALM FTP Indexer** (search.NAPALM FTP Indexer.com) to search for publicly indexed FTP data.

- If found: identify exposed directories or files and explain the risk of misconfigured FTP servers.
- If not found: explain why this is still an important reconnaissance step.

4.7. Task 7 - Pentest Tool Recon Data

Use the **Pentest-Tools.com** platform (or similar passive recon tool) to perform automated passive intelligence collection. Gather:

- Subdomains (active and inactive)
- DNS information
- Technology stack (web frameworks, hosting panels)
- Email patterns / addresses

Explain how this data supports future penetration testing phases (scanning, exploitation, etc.).

4.8. Task 8 - Email Infrastructure Analysis

Based on your research (DNS, MX records, SPF/DMARC):

- Explain the backend procedure of email delivery:
  - Role of **SMTP**
  - Role of **DNS MX records**
- How phishing attackers exploit this infrastructure
- How reconnaissance helps craft convincing phishing campaigns

Relate your explanation to **your target's email configuration**.

4.9. Task 9 - Mapping to Cyber Kill Chain

Map **all your findings** to the **Reconnaissance phase** of the Cyber Kill Chain. Then explain:

- What an attacker would do **next** using your collected data (Weaponization, Delivery, Exploitation, etc.).
- Which attack types could follow (Phishing, Port Scanning, Vulnerability Exploitation, etc.).

4.10. Task 10 - Security Perspective (Defensive Thinking)

From a defender's perspective:

- Explain how organizations can **reduce OSINT exposure**.
- Explain how the **CIA Triad** (Confidentiality, Integrity, Availability) relates to reconnaissance risks.
- Which **controls** (technical, administrative, physical) could prevent intelligence leakage.

5. Deliverables

You must submit a **single comprehensive report** (PDF or DOCX) that includes:

5.1. Methodology Section

For every task (1-10) explain:

- What tool / technique you used
- Why you used it
- Step-by-step process you performed

5.2. Evidence Section (Mandatory)

Each task must include:

- **Screenshots** of your work (terminal, web interface, tool output)
- Clear explanation of each screenshot
- Highlight important findings (IP addresses, ports, subdomains, etc.)

5.3. Findings Summary

Write a concise summary of what intelligence you successfully gathered:

- Technical insights (IPs, open ports, services, subdomains, technologies)
- Organizational structure clues (email patterns, naming conventions, departments)
- Potential risks discovered (missing DMARC, exposed directories, metadata leakage)

5.4. Learning Reflection

Explain:

- What you learned from this assignment
- How reconnaissance builds the foundation of penetration testing
- How you will use these skills as a cybersecurity professional

5.5. Ethical Statement

Add a short note confirming:

- Only **passive** techniques were used
- No intrusion or unauthorized testing was performed
- All findings are based on **publicly available** information

6. Rules & Legality

- **Active scanning is strictly prohibited** (no Nmap, no port scanners, no vulnerability scanners against live systems).

- **No hacking attempts** - do not try to log in, exploit, or modify any data.
- **Only OSINT-based research** using search engines, public databases, & passive tools.
- **Plagiarism will result in rejection** - all work must be your own.
- **Professional formatting** is required (headings, page numbers, clear sections).

By completing this assignment, students will:

- Think like real penetration testers
- Understand how attackers collect intelligence
- Learn how small public data leaks can lead to major breaches
- Gain hands-on OSINT investigation experience

7. Ethical Disclaimer

This assignment is developed strictly for **educational purposes**, **defensive security training**, and **authorized testing environments**. Any misuse of the provided knowledge or techniques outside of legal authorization is strictly prohibited. The student and ITSOLERA Pvt. Ltd. are not liable for any misuse of the techniques demonstrated in this report.

8. References

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